



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
AI-Enabled DevSecOps Framework for Adaptive Quantum-Resilient Software Systems		
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Abstract

Modern apps are usually built from many small connected pieces spread across different systems, which makes them powerful but harder to keep secure — and soon, quantum computers will be able to break the encryption methods we currently rely on to protect data. This project proposes a smart security system that uses AI to constantly watch software for suspicious activity, judge how risky a situation is, and then automatically apply the right level of protection — using regular encryption, quantum-safe encryption, or a mix of both, so companies can upgrade gradually instead of rebuilding everything at once. The system is designed to fit naturally into how modern software teams already work, using popular tools like Git, GitHub Actions, Docker, Kubernetes, SonarQube, OWASP ZAP, Trivy, Prometheus, and Grafana to track code, automate testing, package and run applications, scan for vulnerabilities, and monitor everything after launch. The project will then test how well this system detects threats, how much it slows things down, and how effective it is overall — with the ultimate goal of showing that AI-powered, self-adjusting security can help software stay safe today while also preparing it for the coming era of quantum computing.

Signature of the Guide

HOD